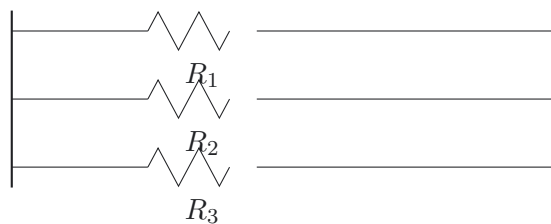


Friction, or Opportunity?

Name: _____

Resistance and conductance describe exactly the same wire, yet hand the imagination two different objects to hold onto.



The question. Which quantity actually measures how easily current flows: resistance, or its reciprocal?

Working rule: Neither quantity is assumed to be the more fundamental one. The task is to notice which is the more natural language for a given circuit.

1. Two words, one wire

define conductance

$$G = \frac{1}{R}$$

Rewrite Ohm's law using G instead of R .

$$V = IR \quad \Longleftrightarrow \quad I = \underline{\hspace{2cm}}$$

2. Two mental models

friction or opportunity

resistance _____

conductance _____

3. Two paths side by side

the resistors above, in parallel

Using $R_1 = 20\ \Omega$, $R_2 = 35\ \Omega$, $R_3 = 60\ \Omega$, complete both routes to the same answer.

The resistance-first route	The conductance-first route
$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$	$G = G_1 + G_2 + G_3$
$R =$	$R = 1/G =$

Do the two routes agree? They should: explain why they must, using nothing but algebra.

4. The formula, unmasked

the same statement, two languages

$$\frac{1}{R_{\text{total}}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} \quad \equiv \quad G_{\text{total}} = G_1 + G_2 + G_3$$

Which of these two, written cold with no context, looks more like it needs explaining? Why?

5. The other direction

series

Take the same three resistors end to end instead of side by side.

Arrangement	Adds simply
Series	_____
Parallel	_____

Complete the rule of thumb in one sentence: resistance is the natural language of _____ circuits; conductance is the natural language of _____ circuits.

6. Core results

what this enquiry actually earned

The relations established along the way.

$$G = \frac{1}{R} \quad G_{\text{total}} = G_1 + G_2 + G_3 \text{ (parallel)} \quad R_{\text{total}} = R_1 + R_2 + R_3 \text{ (series)}$$

Concepts earned, in order.

- Resistance and conductance are the same physics, described by two different mental models: obstacle versus capacity.
- The “mysterious” parallel-resistor formula is exactly what “capacities add” looks like once rewritten in terms of resistance.
- Neither quantity is more fundamental; each is the natural language of a different circuit arrangement.

Every substantial claim should be accompanied by a reason, calculation, or counterexample.